

## 1. TF

11 columns from 프로젝트1. 힌트 파일\_chip 양불 Data - Sheet1 (2)

| lot_id | wafer_id | item_id | subitem_id | chip_x | chip_y | BIN | Defect | CD     | THK    | 양/불 판정 |
|--------|----------|---------|------------|--------|--------|-----|--------|--------|--------|--------|
| AAA.1  | 1        | BIN     | 1          | -1     | 3      | 2   | 34.00  |        |        | 양호     |
| AAA.1  | 1        | BIN     | 2          | 0      | 3      | 1   | 32.00  |        |        | 양호     |
| AAA.1  | 1        | BIN     | 3          | 1      | 3      | 1   | 30.00  |        |        | 양호     |
| AAA.1  | 1        | BIN     | 4          | -3     | 2      | 1   | 33.00  | 114.81 | 160.35 | 양호     |
| AAA.1  | 1        | BIN     | 5          | -2     | 2      | 2   | 25.00  |        |        | 양호     |
| AAA.1  | 1        | BIN     | 6          | -1     | 2      | 1   | 31.00  |        |        | 양호     |
| AAA.1  | 1        | BIN     | 7          | 0      | 2      | 1   | 31.00  | 110.27 | 155.12 | 양호     |
| AAA.1  | 1        | BIN     | 8          | 1      | 2      | 1   | 29.00  |        |        | 양호     |
| AAA.1  | 1        | BIN     | 9          | 2      | 2      | 1   | 29.00  |        |        | 양호     |
| AAA.1  | 1        | BIN     | 10         | 3      | 2      | 1   | 27.00  | 115.15 | 153.20 | 양호     |
| AAA.1  | 1        | BIN     | 11         | -3     | 1      | 1   | 33.00  |        |        | 양호     |
| AAA.1  | 1        | BIN     | 12         | -2     | 1      | 1   | 33.00  | 112.47 | 154.51 | 양호     |
| AAA.1  | 1        | BIN     | 13         | -1     | 1      | 1   | 31.00  |        |        | 양호     |
| AAA.1  | 1        | BIN     | 14         | 0      | 1      | 1   | 34.00  | 109.26 | 150.94 | 양호     |
| AAA.1  | 1        | BIN     | 15         | 1      | 1      | 1   | 30.00  |        |        | 양호     |
| AAA.1  | 1        | BIN     | 16         | 2      | 1      | 2   | 30.00  | 113.75 | 156.48 | 양호     |
| AAA.1  | 1        | BIN     | 17         | 3      | 1      | 1   | 33.00  |        |        | 양호     |
| AAA.1  | 1        | BIN     | 18         | -4     | 0      | 1   | 28.00  |        |        | 양호     |
| AAA.1  | 1        | BIN     | 19         | -3     | 0      | 1   | 31.00  | 115.80 | 152.92 | 양호     |
| AAA.1  | 1        | BIN     | 20         | -2     | 0      | 1   | 27.00  |        |        | 양호     |
| AAA.1  | 1        | BIN     | 21         | -1     | 0      | 1   | 32.00  | 109.93 | 151.21 | 양호     |
| AAA.1  | 1        | BIN     | 22         | 0      | 0      | 1   | 29.00  | 109.63 | 150.25 | 양호     |
| AAA.1  | 1        | BIN     | 23         | 1      | 0      | 1   | 32.00  | 110.29 | 151.84 | 양호     |
| AAA.1  | 1        | BIN     | 24         | 2      | 0      | 1   | 27.00  |        |        | 양호     |
| AAA.1  | 1        | BIN     | 25         | 3      | 0      | 3   | 27.00  | 107.31 | 176.39 | 불량     |
| AAA.1  | 1        | BIN     | 26         | 4      | 0      | 1   | 28.00  |        |        | 양호     |
| AAA.1  | 1        | BIN     | 27         | -3     | -1     | 1   | 27.00  |        |        | 양호     |
| AAA.1  | 1        | BIN     | 28         | -2     | -1     | 2   | 28.00  | 113.35 | 147.82 | 양호     |
| AAA.1  | 1        | BIN     | 29         | -1     | -1     | 5   | 25.00  |        |        | 불량     |
| AAA.1  | 1        | BIN     | 30         | 0      | -1     | 1   | 27.00  | 110.08 | 149.11 | 양호     |
| AAA.1  | 1        | BIN     | 31         | 1      | -1     | 1   | 34.00  |        |        | 양호     |
| AAA.1  | 1        | BIN     | 32         | 2      | -1     | 2   | 30.00  | 109.95 | 156.83 | 양호     |
| AAA.1  | 1        | BIN     | 33         | 3      | -1     | 1   | 30.00  |        |        | 양호     |
| AAA.1  | 1        | BIN     | 34         | -3     | -2     | 1   | 33.00  | 114.94 | 149.05 | 양호     |

Data table:  
프로젝트1. 힌트 파...

## 2. Yield

### 프로젝트1. 힌트 파일\_chip 양불 Data\_수율 - Sheet1

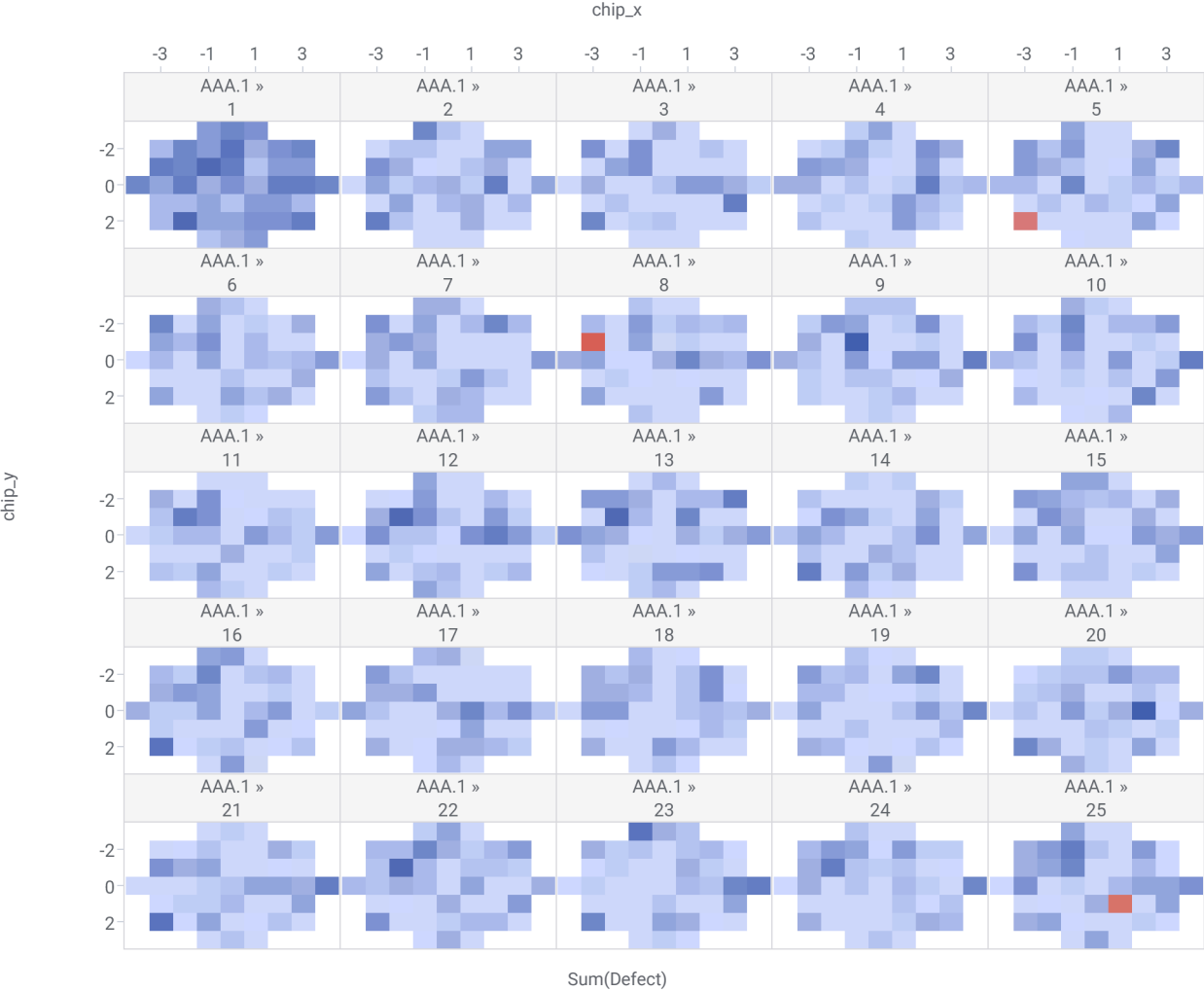
| lot_id | wafer_id | 양호한 칩 개수 | Sum(Defect) | Avg(CD) | Avg(THK) | 수율    |
|--------|----------|----------|-------------|---------|----------|-------|
| AAA.1  | 1        | 40       | 1280.00     | 111.96  | 154.43   | 93.02 |
| AAA.1  | 2        | 38       | 1511.00     | 116.71  | 155.01   | 88.37 |
| AAA.1  | 3        | 39       | 1561.00     | 115.69  | 153.72   | 90.70 |
| AAA.1  | 4        | 38       | 1506.00     | 115.69  | 153.72   | 88.37 |
| AAA.1  | 5        | 37       | 1737.03     | 116.71  | 153.72   | 86.05 |
| AAA.1  | 6        | 39       | 1520.00     | 116.71  | 153.72   | 90.70 |
| AAA.1  | 7        | 39       | 1513.00     | 115.69  | 153.72   | 90.70 |
| AAA.1  | 8        | 36       | 1814.51     | 115.69  | 153.72   | 83.72 |
| AAA.1  | 9        | 41       | 1493.00     | 116.71  | 153.72   | 95.35 |
| AAA.1  | 10       | 40       | 1516.00     | 116.71  | 153.72   | 93.02 |
| AAA.1  | 11       | 40       | 1536.00     | 115.69  | 153.72   | 93.02 |
| AAA.1  | 12       | 40       | 1504.00     | 115.69  | 153.72   | 93.02 |
| AAA.1  | 13       | 39       | 1472.00     | 115.69  | 155.01   | 90.70 |
| AAA.1  | 14       | 40       | 1521.00     | 115.69  | 153.72   | 93.02 |
| AAA.1  | 15       | 39       | 1518.00     | 116.71  | 155.01   | 90.70 |
| AAA.1  | 16       | 41       | 1522.00     | 115.69  | 153.72   | 95.35 |
| AAA.1  | 17       | 40       | 1526.00     | 115.69  | 153.72   | 93.02 |
| AAA.1  | 18       | 38       | 1514.00     | 115.69  | 154.67   | 88.37 |
| AAA.1  | 19       | 37       | 1538.00     | 115.69  | 154.67   | 86.05 |
| AAA.1  | 20       | 38       | 1494.00     | 115.69  | 153.72   | 88.37 |
| AAA.1  | 21       | 39       | 1527.00     | 115.69  | 153.72   | 90.70 |
| AAA.1  | 22       | 39       | 1488.00     | 116.71  | 154.67   | 90.70 |
| AAA.1  | 23       | 39       | 1534.00     | 116.71  | 155.01   | 90.70 |
| AAA.1  | 24       | 40       | 1518.00     | 116.71  | 153.72   | 93.02 |
| AAA.1  | 25       | 38       | 1756.97     | 116.71  | 153.72   | 88.37 |
| AAB.1  | 1        | 40       | 1501.00     | 115.69  | 153.72   | 93.02 |
| AAB.1  | 2        | 40       | 1520.00     | 114.73  | 159.93   | 93.02 |
| AAB.1  | 3        | 38       | 1774.11     | 113.72  | 159.93   | 88.37 |
| AAB.1  | 4        | 39       | 1517.00     | 113.72  | 160.91   | 90.70 |
| AAB.1  | 5        | 37       | 1536.00     | 113.72  | 160.91   | 86.05 |
| AAB.1  | 6        | 38       | 1505.00     | 114.73  | 161.65   | 88.37 |
| AAB.1  | 7        | 38       | 1507.00     | 113.72  | 159.93   | 88.37 |
| AAB.1  | 8        | 38       | 1787.09     | 113.72  | 159.93   | 88.37 |
| AAB.1  | 9        | 41       | 1477.00     | 113.72  | 159.93   | 95.35 |

Data table:

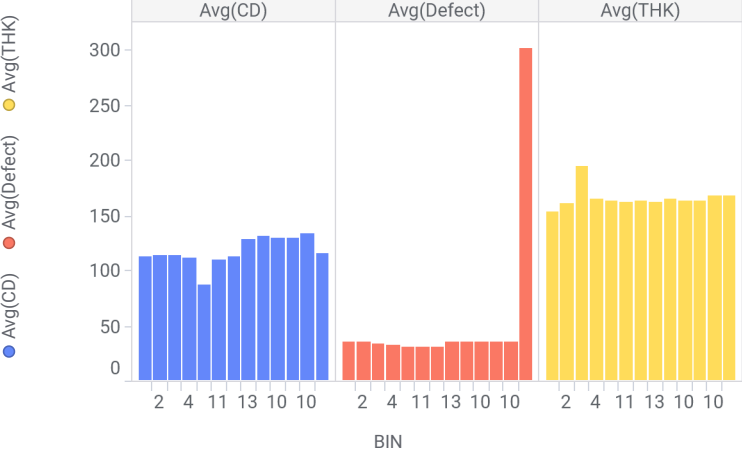
프로젝트1. 힌트 파...

3. Visualization

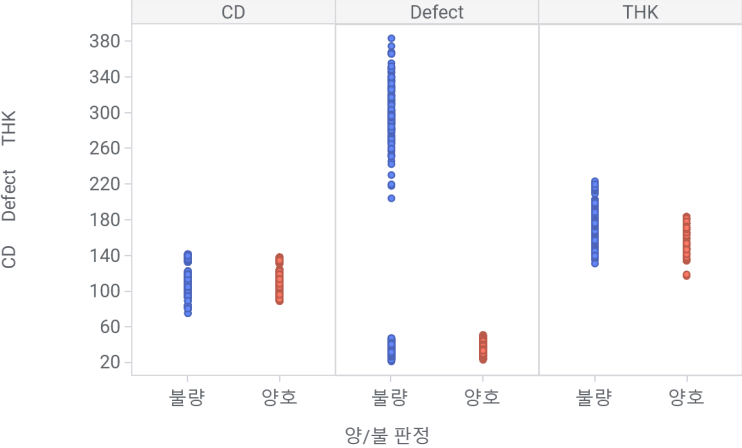
Defect per chip\_x and chip\_y



CD, Defect, THK per BIN

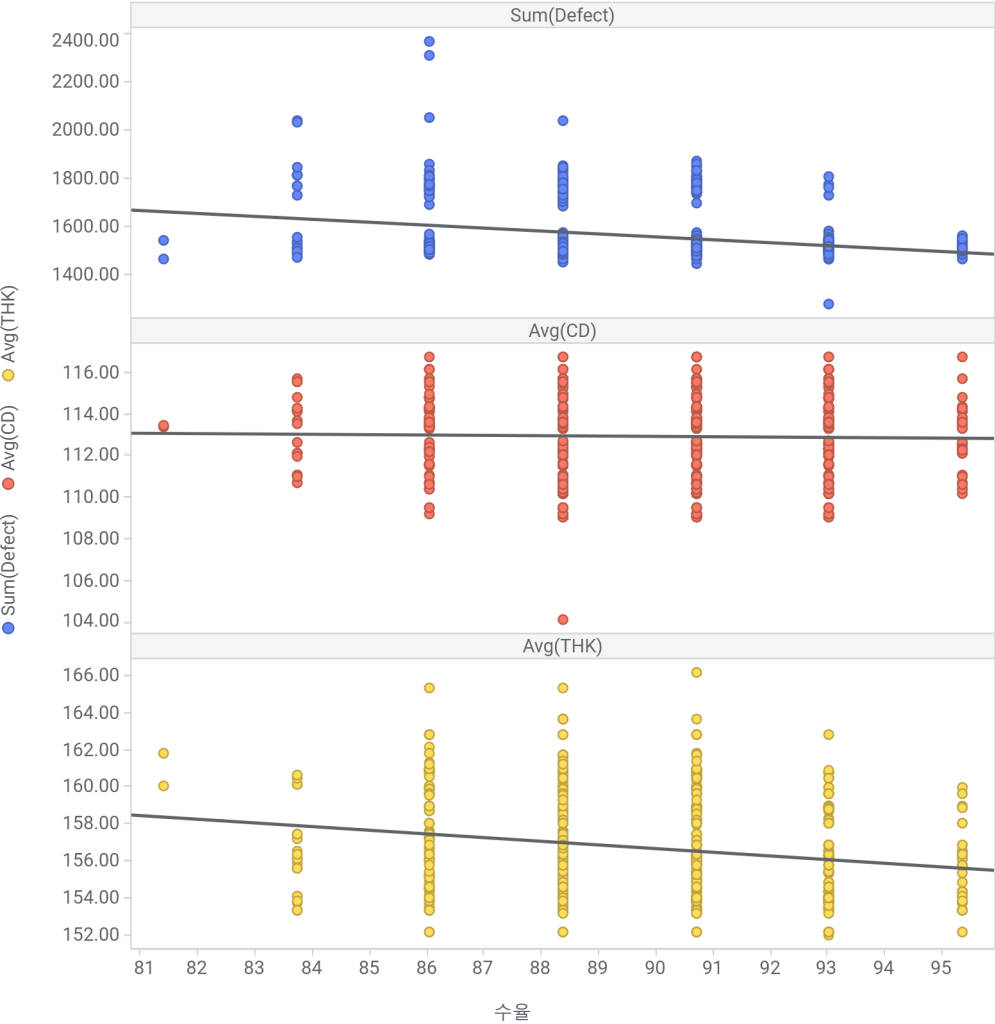


CD, Defect, THK vs. 양/불 판정



4. Visualization

Sum(Defect), Avg(CD), Avg(THK) vs. 수율



Data Relationships (3) (Linear Regression)

| Y (numerical) | X (numerical) | p-value ▲ | RSq ▼ | FStat  | R     |
|---------------|---------------|-----------|-------|--------|-------|
| 수율            | Sum(Defect)   | 6.93E-023 | 0.09  | 101.92 | -0.30 |
| 수율            | Avg(THK)      | 6.77E-010 | 0.04  | 38.84  | -0.19 |
| 수율            | Avg(CD)       | 4.56E-001 | 0.00  | 0.56   | -0.02 |

Data Relationships (3) (Details)

